

Biologically unavoidable sequences: a classification and related notes

10 September 2026 — research packet for discussion

The main claim

Under Definition 1 of Alexander’s *Biologically unavoidable sequences* [1], the biologically unavoidable sequences over a finite alphabet are **exactly the eventually periodic sequences**. The new direction is an explicit avoiding population for every other sequence. The eventually periodic direction is Theorem 10 of the published paper (Theorem 6 of the nine-page arXiv version), with the small proof repair below.

This packet presents a proof for scrutiny. The negative construction has substantial Lean verification, described in the last section. Neither independent human review nor literature priority is claimed.

We use zero-based indices and edge labels. A population is an infinite directed graph with finitely many roots, finite outdegree at every vertex, real birthdates strictly increasing along edges and finite sublevel sets, and an incoming edge of each alphabet label at every non-root. A realizing path may start at **any** vertex.

1. The binary construction and proof

Fix $s : \mathbb{N} \rightarrow \{0, 1\}$ and put

$$r(2j) = s(j), \quad r(2j + 1) = 1 - s(j).$$

Let P_s have vertices $\mathbb{N}$ and birthdate $t(v) = v$. For every $v \geq 2$ put the following two incoming edges, and no others:

$$\begin{aligned} v - 1 &\longrightarrow v \quad \text{with label } r(v), \\ v - 2 &\longrightarrow v \quad \text{with label } 1 - r(v). \end{aligned}$$

There is no edge $0 \rightarrow 1$. The roots are exactly 0 and 1. Vertex 0 has one child, and each vertex $u \geq 1$ has two children. Every non-root has two distinct parents, with complementary labels. Birthdates increase strictly, their sublevel sets are finite, and the graph is infinite. Thus these are populations in the paper’s precise edge-labelled sense.

Claim. If P_s realizes s , then s is eventually periodic.

Suppose $v_0, v_1, \dots$ is an infinite matching path. Each increment is 1 or 2. Inductively,

$$v_k \geq 2k.$$

Indeed, if $v_k \geq 2k + 1$, the next vertex is at least $2k + 2$. If $v_k = 2k$, an increment of 1 would have label $r(2k + 1) = 1 - s(k)$, so is impossible. At $k = 0$ this edge is absent; the same conclusion holds. The base case is $v_0 \geq 0$.

Consequently $d_k = v_k - 2k$ is a nonnegative integer sequence with increments in $\{-1, 0\}$. It eventually stabilizes at some $d \geq 0$. From some K onward every path increment is therefore 2, so

$$s(k) = 1 - r(2k + d + 2), \quad k \geq K.$$

If $d = 2e + 1$, this says $s(k) = s(k + e + 1)$, giving positive eventual period $e + 1$. If $d = 2e$, it says $s(k) = 1 - s(k + e + 1)$; applying the relation twice gives period $2e + 2$. Either case forces eventual periodicity.

Thus **every non-eventually-periodic binary target is avoided by its own P_s , from all starting vertices**. The graph depends on the target; this does not produce one graph avoiding every aperiodic sequence.

2. Finite alphabets and vertex genders

Let A be finite and $s : \mathbb{N} \rightarrow A$ not eventually periodic. At least one indicator projection $\pi : A \rightarrow \{0, 1\}$ has a non-eventually-periodic image $\pi \circ s$. Otherwise all finitely many symbol indicators would be eventually periodic. Beyond the maximum of their starting thresholds, a common multiple of their positive periods would be a period of s .

Construct the binary population $P_{\pi \circ s}$. Replace each vertex v by the copies (v, a) , $a \in A$. For every binary edge $u \rightarrow v$ with label c , add

$$(u, a) \longrightarrow (v, b) \quad \text{with label } a \quad \text{for all } a, b \in A \text{ satisfying } \pi(a) = c.$$

Give each copy birthdate v . There are $2|A|$ roots, at most $2|A|$ children per vertex, and exactly one incoming parent of each label at every non-root. The other axioms are inherited. An s -path would project to a $\pi \circ s$ -path in the binary population, which is impossible.

Moreover, label a is the source copy's second coordinate: assigning gender a to (v, a) makes this a **vertex-gendered** population. For a binary alphabet this yields four roots and at most four children per vertex. The smaller two-root/two-child construction uses the more general edge-labelled definition.

Thue–Morse example. Put $t(k)$ equal to the parity of the number of ones in the binary expansion of k . Then $t(2k) = t(k)$ and $t(2k + 1) = 1 - t(k)$, so the auxiliary sequence r equals t itself. The incoming labels at v are simply $t(v)$ and $1 - t(v)$.

For completeness, t is not eventually periodic. If $p > 0$ were an eventual period, then for every sufficiently large m , $t(2^m - p) = t(2^m) = 1$. But $2^m - p$ is the m -bit complement of $p - 1$, so

$$t(2^m - p) = (m \bmod 2) \text{ xor } t(p - 1),$$

which alternates with m . This is a contradiction.

3. A small repair to the positive direction

Write V_j for the finite set of vertices reachable within j edges of a root. In Theorem 10, deleting V_{k-1} can leave a surviving non-root with only some of its required incoming labels.

For example, take vertices a_i, b_i at integer time $i \geq 0$, with the time-zero pair as roots. Normally every vertex at time $i \geq 1$ has parent a_{i-1} of label 0 and parent b_{i-1} of label 1. Replace the label-0 parent of a_2 by a_0 . Deleting the roots leaves a_2 with a label-1 parent but no label-0 parent.

After deleting V_{k-1} , also delete **all** remaining incoming edges to the surviving boundary $V_k \setminus V_{k-1}$. Those finitely many vertices become roots. Every vertex outside V_k keeps all its original parents: a parent in V_{k-1} would put its child in V_k . Hence the modified graph is a population. Apply the periodic-path theorem there, and prepend the prescribed k labels in the original graph. Its starting point lies outside V_{k-1} , so these backward choices cannot reach a root too early. The purely periodic case needs no deletion.

This repairs the proof step and preserves the positive theorem. More generally, prepending any finite word preserves biological unavoidability in both directions.

4. The other questions in the paper

Universal avoiding populations: a scoped negative answer

Fix an avoidable sequence s . Interpret an embedding as an injective vertex map preserving directed edges and their labels. Then there is no universal population avoiding s . In fact, even a countable family of countable finite-outdegree hosts cannot contain every avoiding population under these embeddings.

Here is the argument, using the standard degree-growth diagonal obstruction for locally finite graphs [2]. Fix one avoiding population P . Replace each vertex v by a nonempty finite number $f(v)$ of copies, and replace each edge by all corresponding edges between its copy sets. Birthdates are inherited. All axioms survive, and projection and lifting show that this blowup has exactly the same infinite label language as P .

Choose a directed ray $v_0 \rightarrow v_1 \rightarrow \dots$ in P ; one exists by finite roots, finite outdegree, and König's lemma. Enumerate all pairs of a proposed host and one of its vertices as (W_i, u_i) . Let M_i be the maximum outdegree in the finite forward ball of radius i about u_i . Set $f(v_{i+1}) = M_i + 1$, and take one copy elsewhere, including at v_0 . If an embedding sent v_0 to the enumerated u_i , a chosen copy of v_i would map inside that ball yet have at least $M_i + 1$ distinct children. Injectivity contradicts the definition of M_i .

The hosts need not avoid s . The result also excludes the more restrictive induced or birthdate-preserving embeddings. It says nothing comparable about noninjective homomorphisms, subdivisions, or classes with a fixed source-degree bound. The paper's broader wording does not select one of these conventions, so this is a scoped answer.

Matching-tree ranks: exact barriers, but a coarse ordinal invariant

For a start v , let $T(v, s)$ be the tree of finite paths matching prefixes of s . It is finitely branching. König's lemma gives

$$s \text{ is not realized from } v \iff T(v, s) \text{ has finite height.}$$

Thus P avoids s exactly when each starting vertex has some finite blocking prefix. The bound may depend on the start.

Every finite word occurs somewhere in every population: choose an endpoint outside V_{m-1} for a word of length m , and select parents in reverse label order. Therefore, in an avoiding population, the finite heights are unbounded over all starts. If a super-root is attached above all the $T(v, s)$, its well-founded rank is exactly ω . This gives a precise rank test, but that one aggregate rank cannot distinguish avoiders. It is not a full classification by richer ordinal invariants.

A category consequence

Let K_v be the set of sequences realized from v . Finite branching makes K_v closed: if every finite prefix is realizable from v , König's lemma gives the infinite path. The population is countable by its finite birthdate sublevels, so $L(P) = \bigcup_v K_v$ is F_σ .

If some K_v contains a cylinder of all sequences extending a finite word w , deleting the w part of each realizing path shows that P realizes every sequence. Thus either P is universal, or every K_v is nowhere dense, $L(P)$ is meagre, and the avoided language is a dense G_δ .

5. Why individual avoidance is compatible with many aperiodic paths

Every population has a root-starting path with labels in any prescribed nonempty one-sided subshift X (a closed, forward-shift-invariant set).

Choose $s \in X$. For arbitrarily large N , choose an endpoint outside V_{N-1} . Its ancestor set is finite, so take a prefix of s longer than any path ending there. Prescribe that prefix backwards until a root is reached. Reading forward gives a factor of s , on a root path of length at least N . The forest of such factor-labelled root paths is prefix closed, finitely branching and has finitely many roots. König's lemma gives a path whose every prefix is a factor of s , hence whose labels lie in X .

Apply this to the orbit closure of a mechanical word $s_\alpha(k) = \lfloor (k+1)\alpha \rfloor - \lfloor k\alpha \rfloor$. Every block of length m has a number of ones differing from $m\alpha$ by less than one. The root path in that orbit closure therefore has frequency α . For irrational α it is aperiodic, and different slopes give different sequences. **Every binary population realizes continuum many aperiodic sequences**, despite every individual aperiodic sequence having an avoiding population.

6. Universality and finite observation in the new family

The construction has a second exact property:

$$P_s \text{ realizes every binary sequence} \iff s \text{ is eventually periodic.}$$

The negative direction is §1. If s has eventual period p , then r has eventual period $q = 2p$. Form a finite residue multigraph modulo q , with the two incoming transitions into residue a , labelled by the eventual $r(a)$ and its complement, and marked with delays 1 and 2. Every state has an incoming transition of each label. Every finite word can therefore be read backwards; finite branching and finitely many possible starts give an infinite path for every infinite word. Starting sufficiently late in P_s , lift each transition by its marked delay. This realizes the word in the actual population. The marked delays retain the information needed to lift the quotient path.

The graph through vertex N depends only on $s(0), \dots, s(\lfloor N/2 \rfloor)$. Any such prefix can be continued with zeros or with Thue–Morse. The resulting populations agree on that entire finite region, but one is universal and the other is not. This also obstructs any finite adaptive sequence of local edge queries.

Even a program describing the graph does not decide universality. Given a machine M , let $s_M(k)$ be the k th Thue–Morse bit if M has not halted within k steps, and zero otherwise. This sequence and its graph are total computable objects. The graph is universal exactly when M halts. Conversely, using zeros until a halt is detected and then starting a Thue–Morse tail gives a universal graph exactly when M never halts. Thus neither universality nor non-universality is semidecidable on these presentations, even with the promise that all population axioms hold.

For a fixed aperiodic s , breadth-first matching from any vertex terminates with an empty set by §1 and König’s lemma. There is no bound uniform in the target even for starting vertex 1: a target with $N + 1$ initial zeros and an aperiodic tail admits N matching edges along $v_k = 1 + 2k$. Obtaining useful explicit bounds for the fixed Thue–Morse target was left as a quantitative question in the September 10 packet. **September 20/21 update:** the separate complement-and-barriers addendum supplies an $O((i + 1) \log(i + 2))$ cutoff from start i ; sharp growth remains open. The countable-avoidance addendum also proves that any prescribed countable family of aperiodic targets can be avoided simultaneously, with optimal uniform outdegree for every finite alphabet. These new hand proofs are not included in the older Lean coverage described below.

Verification and questions for discussion

The included Lean source checks the binary path argument, the indicator reduction, and all population axioms for the finite-alphabet construction using natural-valued birthdates. Embedding those dates in the reals is a hand argument in this snapshot: for any real R , choose a natural $T > R$ and use the finite strict sublevel below T . The positive theorem, Thue–Morse aperiodicity, and the other extensions above are hand proofs.

The recorded Lean 4.32.0-rc1 check exits successfully; its printed endpoints use only `propext`, `Classical.choice`, and `Quot.sound`. The finite Python controls independently transcribe edges and matching sets. They supplement the proofs, and do not establish infinite claims by extrapolation.

The user selected the paper and directed the research. OpenAI Codex developed the construction, proofs, formalization and exposition. No independent human mathematical review is recorded. We would especially welcome corrections to the definition correspondence, pointers to earlier solutions, and clarification of the intended universal-object notion. The manuscript’s final classification question is answered by the proposed proof; its full range of ordinal or embedding questions is not claimed to be exhausted.

References

1. Samuel A. Alexander, *Biologically unavoidable sequences*, Electronic Journal of Combinatorics **20**(1) (2013), P31. Published paper, DOI, arXiv:1212.0186. We refer to the published Definition 1, Theorem 10 and §6.
2. Florian Lehner, *A note on classes of subgraphs of locally finite graphs*, author-hosted primary paper. Used for the antecedent degree-growth obstruction; the label-language preserving blowup above is spelled out separately.